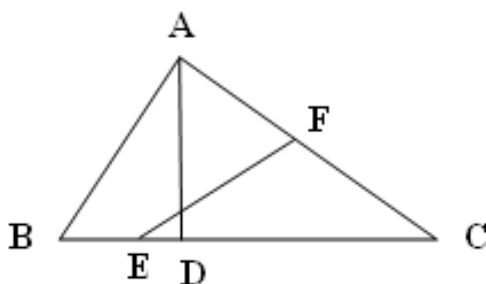
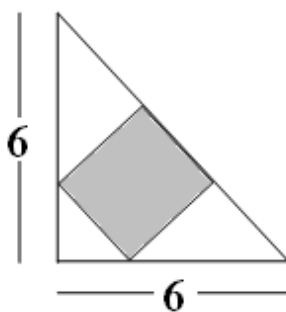


JMO – 2012

1. A manufacturer built a machine which will address 500 envelopes in 8 minutes. He wishes to build another machine so that when both are operating together will address 500 envelopes in 2 minutes. How many minutes would the second machine take to address 500 envelopes alone? [5]
2. Mohan ordered 4 pairs of black socks and some additional pairs of blue socks. The price of the black socks per pair was twice that of blue. When the order was filled, it was found that the number of pairs of the two colors had been interchanged. This increased the bill by 50%. What was the ratio of the number of pairs of black socks to the number of pairs of blue socks in the original order? [5]
3. Ram and Shyam at the same time ride from Bhubaneswar to Puri 60 km away. Ram travels 4 km/hr slower than Shyam reaches Puri and at once turns back meeting Ram 12 km from Puri. What was the speed of Ram? [5]
4. A merchant buys goods at 25% off the list price. He desires to mark the goods so that he can give a discount of 20% on the marked price and still gets a profit of 25% on the selling price. What percent of the list price must he mark on the goods? [5]
5. In $\triangle ABC$, $AD \perp BC$. $BC = 8$ cm and $AD = 6$ cm. If E and F are mid points of BD and AC respectively find EF. [5]



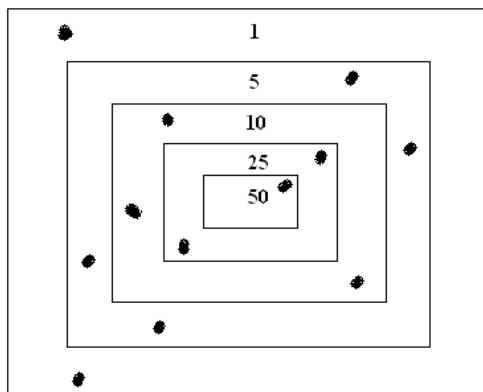
6. A679B is a five digit number where A, B are written for two missing digits of the number. If the number is divisible by 72, then find the value of A and B. [5]
7. (i) Find the area of the shaded square. [3]



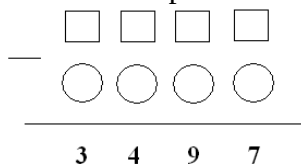
- (ii) Find the last digit of 3^{1486} . [3]
8. Prove that n must be odd for $(1+1/2)(1+1/3)(1+1/4).....(1+1/n)$ to be integer. [4]
9. A piece of string measures $2/3$ m. Without the use of any other measuring tools, how would you use the string to measure $1/2$ m? [5]

10. Two players take it in turns to fire at a target. They get six shots each. Both players scored exactly the same total. If the second player scored 26 points with her first two shots, which player hits the centre?

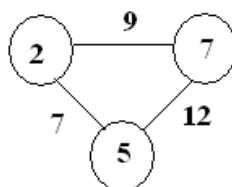
[5]



11. In a particular class, each student reads either mathematics or history; $\frac{1}{4}$ of students reading mathematics and history and $\frac{1}{3}$ of the students reading history read mathematics. What fraction of the class read history?
12. Find the sum of all four digit numbers formed by the digits 1, 2, 3 and 4. Example: the sum of all three digits number formed by digits 1, 2 and 3 is $123+132+213+231+312+321 = 1332$.
13. Mathematician Morgan wrote 1864 "At some point in my life the square of my age was the same as the year." When he was born? For example: if you were 35 years old in the year 1225 then the square of my age at that time is equal to the year $1225 = 35^2$.
14. In the figure, the boxes contain 3, 4, 6 and 9 in some order and the circles contain the digits 4, 5, 8 and 9 in some order. What four digit number is represented by the boxes?

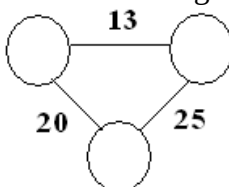


15. Find the value of a if $2^{36} - 1 = 68a19476735$.
16. In rectangle ABCD we have $AD = 1$, P is on AB, and DB and DP trisect ADC. What is the perimeter of DBDP?
17. In the diagram below, the number on the connecting lines is the sum of the two numbers in the circles.



What numbers are missing from the circles in the diagram below?

[5]



JMO QUESTIONS (ORISSA)

18. A draw contains an unsorted collection of black, blue, white and red colored socks. If socks are taken at random, one at a time, what is the minimum number which must be taken to be certain of finding five matching pairs? [5]
19. A bag contains eight yellow marbles, seven red marbles and five black marbles. Without looking in the bag, Deepa removes N marbles all at once. If she is to be sure that, no matter which choice of N marbles she removes, there are at least four marbles of one color and at least three marbles of another colour left in the bag, what is the maximum possible value of N ? [5]

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